

General Resonance Architecture (GRA): From 6200 Daily Thoughts to the Nullification Game

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Abstract

We present a unified mathematical framework — the General Resonance Architecture (GRA) — that models cognitive, social and artificial systems as hierarchical structures of resonant modes and internal contradictions (“foam”). The theory is anchored in the empirical estimate of 6200 thoughts per day (Queen’s University, 2020), each thought representing a micro-decision that shapes subjective reality. GRA introduces a hierarchy of abstraction levels, a foam functional $\Phi(k)$, stability criteria via derivatives with respect to hierarchy, and a nullification operator that resets destabilising resonances. We formalise the hierarchical stability rank R_{hier} and show its application to personal coaching, multi-agent swarms, governance simulation, and even artistic creation. Verification strategies, backend/frontend implementations, and cross-disciplinary applications are outlined.

1 Introduction

The human mind produces a continuous stream of thoughts. Recent work by Queen’s University (2020) estimated this stream at approximately 6200 thoughts per day, each thought corresponding to a transition between mental states. These first reactions to stimuli constitute the raw material from which our subjective reality is constructed. If we can learn to detect, structure and selectively “nullify” patterns in this stream, we can reshape not only behaviour but the very perception of the world.

The General Resonance Architecture (GRA) provides a formal language for this process. Originally developed for reducing “argumentative foam” in

multi-agent debates, GRA has evolved into a broad framework applicable to individuals, organisations, swarms of AI agents, and even physical systems. This paper unifies the key concepts from over 100 GRA repositories into a coherent manifesto.

2 Scientific Anchor: 6200 Thoughts

We adopt the estimate of 6200 thoughts per day as a baseline for the daily flow of micro-decisions. For a subject S , let $M(t)$ denote the mental state at time t , with transitions $\Delta M_i = M(t_{i+1}) - M(t_i)$ for $i = 1, \dots, N$, $N \approx 6200$. Each transition can be annotated with a meta-event m_i : content, emotion, decision.

3 Hierarchical Model and Foam Functional

A system is organised into levels $k = 0, 1, 2, \dots$ representing abstraction/control depth:

- $k = 0$: raw sensations, actions
- $k = 1$: emotions, local plans
- $k = 2$: strategic goals, values
- $k \gg 2$: meta-agents, philosophy, meaning

For each level, define a foam functional $\Phi(k)$ as the expected mismatch, stress, or contradiction at that level. A simple analytic form is:

$$\Phi(k) = Ae^{-\alpha k} + Be^{\beta k}, \quad A, B, \alpha, \beta > 0, \quad (1)$$

where the first term captures stabilisation through abstraction, and the second term captures the risk of over-complexity.

4 Axioms of GRA Nullification

Axiom 1 (Hierarchy). *The system is organised into levels $l = 0, \dots, L$, each with state $x_l \in \mathbb{R}^{n_l}$.*

Axiom 2 (Foam). *For each level, a foam functional $\Phi(x_l) \geq 0$ is defined.*

Axiom 3 (Nullification). *There exists an operation T_l such that $\Phi(T_l(x_l)) \leq \Phi(x_l)$.*

Axiom 4 (Monotonicity). *In the ideal regime, $\Phi_{l+1} \leq \Phi_l$.*

Axiom 5 (Sensitivity). *$\Phi(h)$ is differentiable w.r.t. the hierarchy parameter h ; the sign of $d\Phi/dh$ determines evolution mode (productive collapse, stagnation, degradation).*

5 Stability Criteria

5.1 Continuous Criterion

A nullified state at hierarchy level h^* is *stable* if:

$$\Phi(h^*) = 0, \quad \frac{d\Phi}{dh}(h^*) = 0, \quad \frac{d^2\Phi}{dh^2}(h^*) > 0. \quad (2)$$

This ensures a local minimum of foam.

5.2 Discrete Criterion

For discrete levels l , the stable condition is:

$$\Phi_{l^*} = 0, \quad \Phi_{l^*+1} - \Phi_{l^*} = 0, \quad \Phi_{l^*+1} - 2\Phi_{l^*} + \Phi_{l^*-1} > 0. \quad (3)$$

5.3 Hierarchical Stability Rank R_{hier}

Define the n -th finite difference magnitude:

$$D_n(k) = |\Delta^n \Phi(k)|, \quad \Delta \Phi(k) = \Phi(k+1) - \Phi(k). \quad (4)$$

The rank $R_{\text{hier}} = N$ is the minimal n such that:

- For all $n < N$, $D_n(k)$ is monotonically decreasing for large k (foam damping).
- For $n = N$, $D_N(k)$ eventually stops decreasing and starts to grow or change sign (over-hierarchy).

This rank characterises the system's optimal depth of reflection.

6 Wave–Particle Duality in GRA

Following the multiverse approach, we define a wavefunction over hierarchy levels:

$$\psi(k) \propto \exp(-\lambda\Phi(k)), \quad \lambda > 0. \quad (5)$$

Then $|\psi(k)|^2$ is the probability that the system “lives” at level k . Sharp minima correspond to particle-like stable goals; flat or multi-modal profiles correspond to wave-like wandering.

For two conflicting attractors (slits) x_1^* and x_2^* , with foams $\Phi_1(k)$ and $\Phi_2(k)$, the total amplitude is:

$$A(k) = A_1\psi_1(k) + A_2\psi_2(k)e^{i\varphi(k)}, \quad (6)$$

where $\varphi(k) \sim \Phi_1(k) - \Phi_2(k)$. The interference term:

$$P(k) \propto |A(k)|^2 = |A_1|^2|\psi_1|^2 + |A_2|^2|\psi_2|^2 + 2\Re\{A_1A_2^*\psi_1\psi_2^*e^{i\varphi}\}. \quad (7)$$

A hard observation (projection) collapses the wave to a classical particle choice.

7 Verification Frameworks

7.1 Mathematical Proofs

The stability theorem is proven by showing that the conditions define a strict local minimum of Φ . The swarm subjectivity theorem (GRA-Swarm-Subject) can be formalised using graph connectivity and convexity of the total foam functional.

7.2 Simulation Benchmarks

A benchmark suite (e.g., ‘GRA-Validation-Suite’) generates synthetic data with known ground truth and measures: - Convergence speed to $\Phi \rightarrow 0$ - Recovery time after perturbation - Scalability with number of agents/hierarchy levels

7.3 Empirical Human Studies

Design a mobile diary app (Angel-in-Pocket) to collect thought logs and emotional ratings. Compute $\Phi(k)$ via semantic embeddings and estimate R_{hier} per user. Correlate with standard psychological scales (e.g., BDI, STAI) to validate predictive power.

8 Implementation Blueprint

8.1 Backend (Python + FastAPI)

- Endpoints: ‘/capture’ (thought), ‘/foam’ (profile), ‘/nullify’ (reset), ‘/report’ - PostgreSQL for persistent storage, Redis for caching - GRA core modules packaged as Python wheels

8.2 Frontend (React + React Native)

- Dashboard visualising $\Phi(k)$ as a hierarchical tree - Real-time thought capture via chat interface - Interactive “what-if” simulations (change weights, add new levels)

9 Applications Across Disciplines

9.1 Natural Sciences

- **Physics:** plasma control, quantum analogs (chiral nullification) - **Biology:** immune system modelling (living vaccines), hormonal cycles - **Chemistry:** materials discovery via multiverse hypothesis testing

9.2 Humanities

- **Psychology:** cognitive bias detection and therapy - **Sociology:** policy simulation, conflict resolution - **Linguistics:** semantic clustering, dialogue coherence

9.3 Arts

- **Music:** harmonic progression generation based on resonant modes - **Painting:** colour and composition optimisation - **Literature:** plot development, poetry generation (Ramanujan priors)

10 Conclusion

The GRA framework transforms the chaotic stream of 6200 daily thoughts into a structured hierarchy of resonances and nullifications. By computing the stability rank R_{hier} and applying the nullification operator, individuals, organisations and AI systems can achieve coherent, adaptive states. The

theory is mathematically grounded, verifiable, and implementable, with a wide range of practical applications.

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